

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	15 july 2026
Team ID	-
Project Name	Book Store
Maximum Marks	4 Marks

Functional Requirement:

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration using Name, Email, and Password Input validation for user details Password encryption using bcrypt
FR-2	User Authentication	Secure Login using Email and Password JWT Token Generation User Logout
FR-3	Book Management	View Book Catalog Search Books by Title or Author Filter Books by Category View Book Details
FR-4	Wishlist Management	Add Book to Wishlist Remove Book from Wishlist View Wishlist
FR-5	Shopping Cart	Add Book to Cart Update Book Quantity Remove Book from Cart View Cart Summary
FR-6	Order Management	Place Order View Order History Track Order Status
FR-7	Reviews & Ratings	Submit Review

FR No. Functional Requirement (Epic) Sub Requirement (Story / Sub-Task)

FR-8 Admin Dashboard	Give Book Rating
	View Customer Reviews
	Add/Edit/Delete Books
	Manage Categories
	Manage Users
	Manage Customer Orders
	View Dashboard Statistics

Non-Functional Requirement:

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The application shall provide a simple, responsive, and user-friendly interface that enables users to browse, search, and purchase books easily across desktop and mobile devices.
NFR-2	Security	User passwords shall be encrypted using bcrypt, and JWT-based authentication shall be used to secure user sessions and protected API endpoints.
NFR-3	Reliability	The system shall maintain consistent functionality with accurate processing of book management, shopping cart, orders, and user data without data loss.
NFR-4	Performance	The application shall respond to user requests within a few seconds under normal operating conditions and efficiently handle multiple concurrent users.
NFR-5	Availability	The system shall be available whenever users need to browse books, place orders, or manage inventory, ensuring minimal downtime.
NFR-6	Scalability	The MERN architecture shall support increasing numbers of users, books, and transactions, and allow future integration of payment gateways, cloud deployment, and AI-based recommendation systems.
NFR-7	Maintainability	The application shall follow a modular architecture, making it easier to update, debug, and extend with additional features.
NFR-8	Compatibility	The application shall function correctly on modern web browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari.
NFR-9	Data Integrity	The system shall ensure that all user, book, order, wishlist, and review

NFR No.	Non-Functional Requirement	Description
		data stored in MongoDB remains accurate, consistent, and free from duplication.
NFR- 10	Portability	The application shall be deployable on Windows, Linux, and cloud platforms such as AWS, Azure, or Render with minimal configuration changes.